

Nigel Andati

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EDUCATION

Duke University

Bachelor of Science in Computer Science, Expected May 2028

Durham, NC

EXPERIENCE

Software Engineering Intern

Tech Hub Africa LLC

May 2026 – Present

Accra, Ghana

- Improved page-load performance for AfroEventix's 1,000+ monthly users by shipping 5+ production features and custom REST endpoints (WordPress, JavaScript).
- Developed AI-assisted tooling that converted stakeholder requests into scoped architecture proposals across 3+ client projects.

Founding Engineer (Startup Engineering Lead)

Product Manager Accelerator

Mar 2026 – Present

Remote / Boston, MA

- Owned full-stack engineering for Heiwa, an AI-powered family OS, as the team's sole engineer — architected the technical roadmap end-to-end using React, Supabase, Gemini, and Firecrawl.
- Built logistics parsing from flyers and URLs, and owned calendar sync and OAuth integration from design through production.

Software Engineering Intern

Nexa Consulting

May 2025 – Aug 2025

Nairobi, Kenya

- Designed Python/FastAPI REST endpoints and PostgreSQL data models for a 10K+ user product; added Redis caching and query-index optimizations to reduce p95 latency.
- Implemented JWT authentication and integration tests, raising backend test coverage to 85%+ and reducing production defects.

Full-Stack Software Developer

Body Flex Gyms

May 2024 – Aug 2024

Nairobi, Kenya

- Built and shipped live payment-processing features handling real financial transactions for 200+ monthly customers — early hands-on ownership of a production billing system, improving database reliability and billing-flow stability.

PROJECTS

Market Microstructure Backtesting Engine

Technologies: C++20, Market Microstructure

- Built a C++ backtesting engine that replays order-by-order market data through a custom price-time-priority matching engine, simulating realistic fills, slippage, and market impact for trading strategies; parallelized strategy evaluation across independent simulations with a thread pool, achieving 3.85M events/sec throughput with 1.0 μ s p99 matching latency.

Codewright — Automated Bug Fixing & Feature Generation Engine

Technologies: Python, LLMs, Docker, FastAPI, GitHub API

- Engineered an autonomous coding agent using AST analysis to map repositories, plan patches, and validate changes via parallel test orchestration; achieved 100% patch success rate across 5 repositories.

Semantic Recommendation Algorithm

Technologies: Python, LangChain, OpenAI, ChromaDB

- Built a RAG recommendation service over 7,000+ books using ChromaDB and LangChain, tuning chunking and retrieval.

ACTIVITIES & LEADERSHIP

ColorStack Fellow | CodePath Fellow | Nike Sophomore Recruiting Summit | Global Ambassador, Duke Kunshan University | Secretary, African & Black Students Association | Varsity Soccer | EA Software Engineering Virtual Experience Program

SKILLS

Languages: Python, JavaScript, Java, C++, Go, SQL, HTML/CSS

Frameworks & Libraries: React, Node.js, FastAPI, Flask, Django, Express, Next.js, Redux, Vue.js, LangChain

Databases & Storage: PostgreSQL, MySQL, MongoDB, Redis, Firebase, DynamoDB, Cassandra, SQLite, Elasticsearch, ChromaDB

Cloud & DevOps: AWS (EC2, Lambda, S3), GCP, Azure, Docker, Kubernetes, Terraform, CI/CD, Jenkins, Git/GitHub, Linux

Systems & Architecture: Microservices, REST/GraphQL APIs, WebSockets, gRPC, Kafka, RabbitMQ, System Design

AI & Data: LLMs, RAG, AI Agents, Prompt Engineering, Generative AI, OpenAI

Testing & Tools: JUnit, PyTest, Jest, Selenium, Postman, JSON, YAML, VS Code